

EXPERIMENT A5: VARIABLE INERTIA BAR

REG. No. F19/6452/2021

NAME MBITHI JACKSON NDUVA

DATE: 9/12/2021

PARTNERS

REG. No.

NAME

1. F19/2342/2021

SEAN MICHAEL

2. F19/2362/2021

EDWIN NJERU

3.

4.

OBSERVATIONS AND DATA ANALYSIS

Results

Worksheet 5-1

Weight W ±(N)	Time (t) ±(s)	$t^2 \pm \dots(s)$	$1/t^2$ $\pm \dots(s^{-2}) \times 10^{-2}$
6.730	5.19	26.94	3.712
5.761	5.34	28.52	3.507
4.808	6.25	39.06	2.560
3.898	7.53	56.70	1.764
2.914	8.86	78.50	1.274
1.943	10.18	103.63	0.9649

Analysis:

- Plot a graph W against $1/t^2$
- Plot a graph of t^2 against k .

- Comment on your two graphs and state whether or not the objectives of the experiments have been achieved.

Graph of W against $1/t^2$

Time taken for the mass to strike the floor depends on distance k Weight W provided k is constant. It increases with decrease in mass.

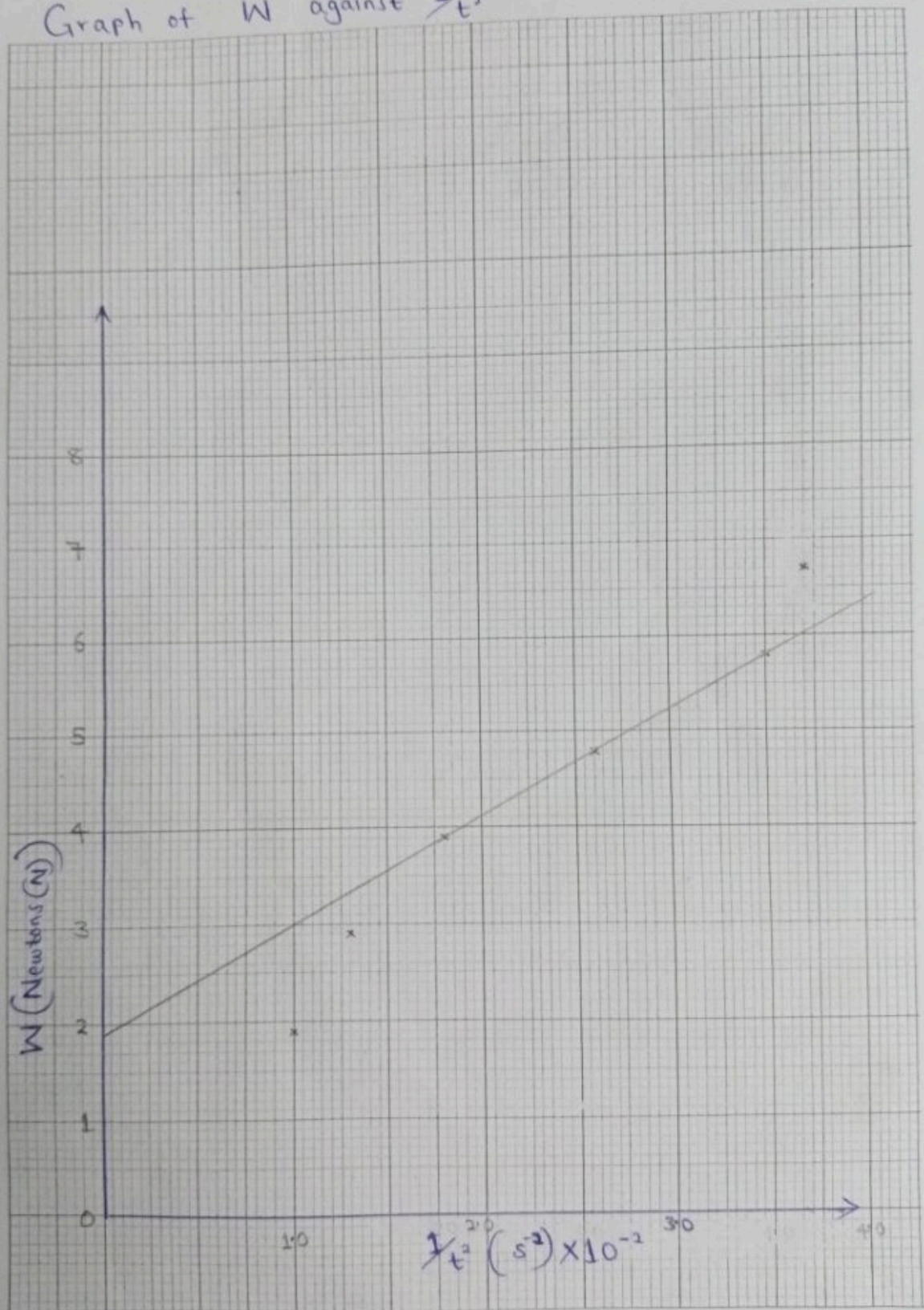
Graph of t^2 against k .

Time taken for the mass to strike the floor is independent of the distance k .

4. Assume no error in W . What other errors are found in this experiment?

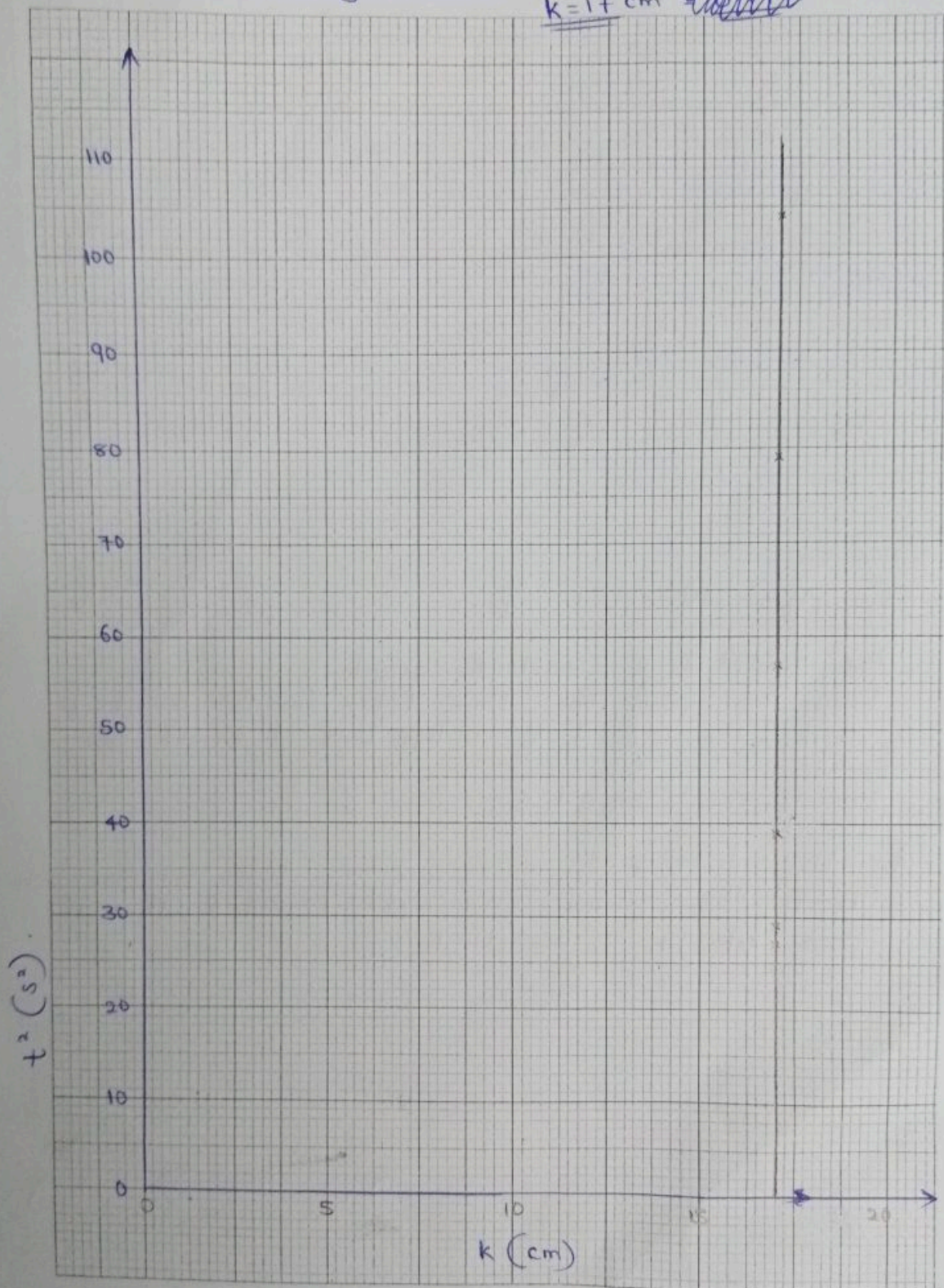
- i) The weight of the thread is not accounted for.
- ii) One may be slow or fast in starting/stopping of the stop watch.
- iii) There is effect of air resistance.

Graph of W against $\frac{1}{t^2}$



Graph of t^2 against k .

$k = 17 \text{ cm}$ ~~$k = 17 \text{ cm}$~~



Worksheet 5-2

Time (t) ±(s)	$t^2 \pm \dots\dots(s^2)$	Distance k ±.....	$\frac{1}{t^2} \times 10^{-2} (s^{-2})$
6.09	37.09	12.5	2.70
7.50	56.25	15.0	1.78
8.05	64.80	17.0	1.54
8.59	73.79	20.0	1.36
8.91	79.39	22.0	1.26
10.12	102.41	27.0	0.976

Analysis

5. Plot a graph of W against $1/t^2$ including error bars on both W and $1/t^2$
6. Plot a graph of t^2 against k including error bars on t^2 .

Conclusions

Graph of W against $\frac{1}{t^2}$

The graph is a constant graph.

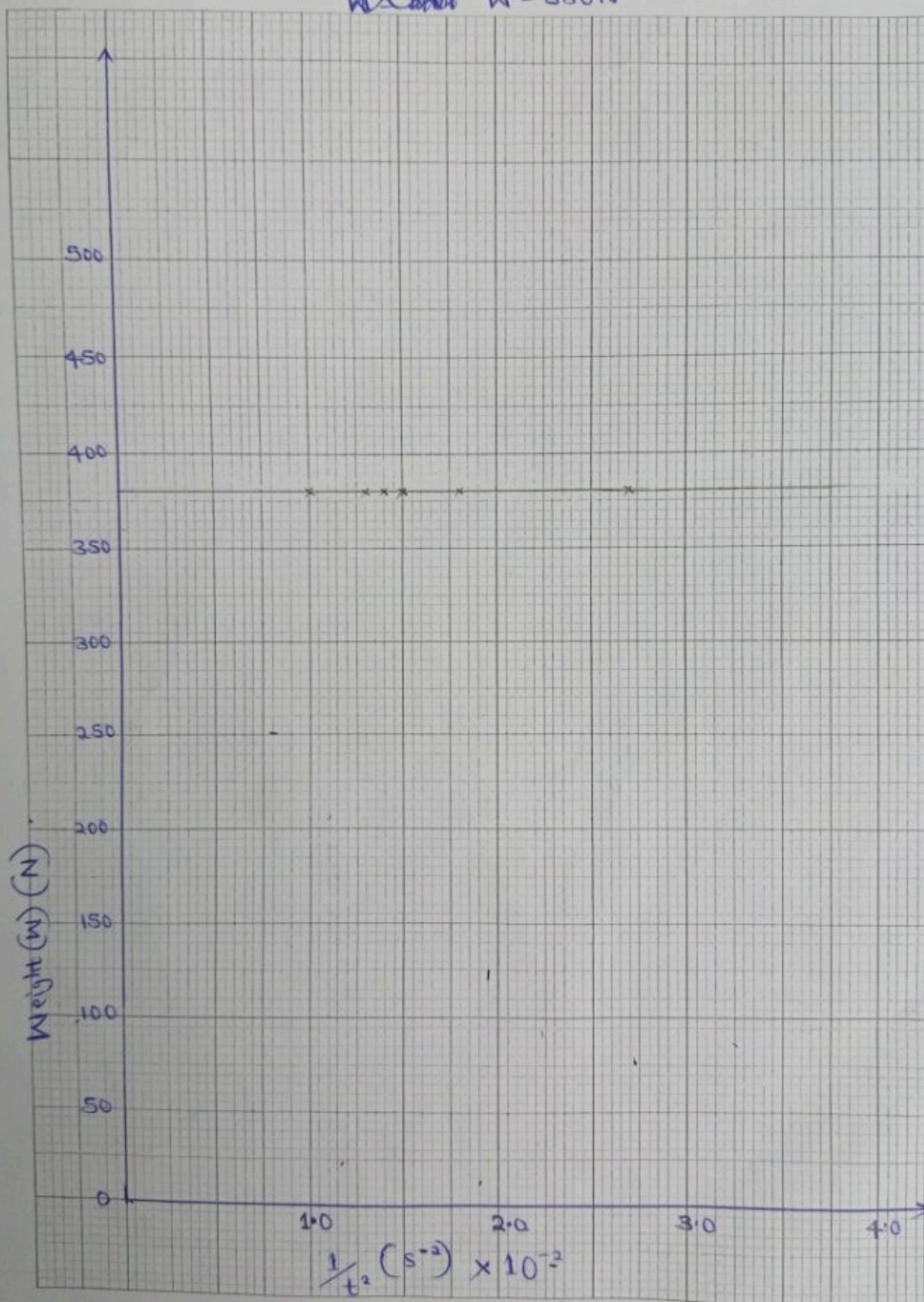
- From the graph, the time taken for the masses to strike the ground is independent of Weight (W) -

Graph of t^2 against k:

As the distance k increases, time taken for masses to strike the ground increases, ~~it~~ provided that the ~~the~~ Weight is constant.

Graph of W against $\frac{1}{t^2}$ Including error bars on both W & $\frac{1}{t^2}$

~~W = 380 N~~ $W = 380 \text{ N}$



Graph of t^2 against k including error bars on t^2

